Google docs link: Biodome facilitator guide

The biodome is a 6 room digital escape room where players have to restart a research station's life support by proving they understand photosynthesis, cellular respiration, and how the two connect.

Estimated play time is around 25 minutes.

1. Running the game

Before class, download the Biodome_Six_Escape_Room.html file and double click it to run in a browser. This is basically a static website so it's only hostable from the file. You can find all my code here and version history: https://github.com/eric-bodhi/biodome

During a game

1. Players type their name in and hit the "enter the dome" button. This also starts the game timer.
2. All the rooms unlock in order with multiple questions labeled A-D.
3. Players have to complete every question, then hit "Try the lock". The game will tell you which questions are wrong and how many are wrong, but it will not show you the answer.
4. Every solved room unlocks a data-log entry that summarizes the big picture. This is good because students can review as they go.
5. *There is also a skip room button so someone can go through the game quickly to review.*

The atmosphere meter is like a grade but the player will never get kicked out of the game. Every wrong answer costs 3% and a hint costs 5%. There are two hints per room.

After playing, the final screen will show a mission report showing: time, atmosphere left (remember just grade), hints, wrong answers, and the 6 data log entries, the player's written explanation for the last question (there is a grading guide at the end of this for that), and all the works cited for the project.

Player directions (read aloud): You are locked in Biodome Six. Solve all six rooms to reopen the airlock. Tap a molecule then a blank to place it, use the menus, buttons, and number dials to answer the rest. When every part is done, try the lock. Wrong tries and hints drain the atmosphere, so check your work first!

2. Room Answer keys

Room 1: Command Deck

HELIX's message: Crew member, the dome's CO2 is climbing and its O2 is falling. When the Life support model crashed, I sealed every door. Rebuild the two master equations so I can reload it, then confirm you know what each process is for.

A. Rebuild the master equations

“Tap a molecule, then tap a blank to place it. Tap a filled blank to clear it. Every molecule can be used in both equations, and one tile is to confuse players. Order does not matter.”

Answer:

Photosynthesis: $6CO_2 + 6H_2O + \text{Light energy} \rightarrow C_6H_{12}O_6 + 6O_2$

Cellular respiration: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{Energy (ATP + heat)}$

Why: Photosynthesis: $6CO_2 + 6H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6O_2$. Cellular respiration: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy (ATP + heat)}$. Chlorophyll is the decoy: it absorbs light

but is not consumed, so it is not a reactant.

B. Purpose check: photosynthesis

“What is the main purpose of photosynthesis for the plant?”

Answer: B. To convert light energy into chemical energy stored in the bonds of glucose

Why: Photosynthesis is anabolic and endergonic: it captures light energy and stores it as chemical energy in glucose. O_2 is a byproduct of splitting water, not the goal (A). C describes cellular respiration; D reverses the energy conversion.

C. Purpose check: cellular respiration

“What is the main purpose of cellular respiration?”

Answer: D. To release the energy stored in glucose and capture part of it in ATP, the molecule cells spend to do work

Why: Cellular respiration is catabolic and exergonic: it oxidizes glucose and transfers part of the released energy to ATP. Energy is never created (A, first law of thermodynamics); CO_2 is a waste product, not the purpose (B); C describes photosynthesis.

Hints (each costs 5% atmosphere):

(1) In photosynthesis the plant takes in a gas, a liquid, and energy. In cellular respiration a cell takes in a sugar and a gas. Chlorophyll absorbs light but isn’t used up, so it isn’t a reactant.

(2) The two equations are mirror images: the reactants of photosynthesis are the products of respiration. O_2 is released by photosynthesis, but it is a byproduct, not the goal.

Data log entry unlocked: The equations are mirror images

Photosynthesis, $6CO_2 + 6H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6O_2$, is **anabolic and endergonic**: it stores light energy in the bonds of glucose.

Cellular respiration, $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$, is **catabolic and exergonic** ($\Delta G \approx -2{,}870$ kJ per mole of glucose): it releases that stored energy and captures about a third of it in ATP. The rest leaves as heat.

Room 2: Navigation Bay

Where each stage happens. Location: Chloroplast, cytoplasm, mitochondrion.

HELIX’s message: The repair drones need coordinates. Assign every stage to the compartment where it happens, or they’ll weld the wrong membrane. The station map is below.

A. Assign each stage a location

“Use the station map. Two stages share a location, and some locations are decoys.”

Answer:

Light-dependent reactions: Thylakoid membranes (chloroplast)
Calvin cycle: Stroma (chloroplast)
Glycolysis: Cytoplasm (cytosol)
Pyruvate oxidation: Mitochondrial matrix
Citric acid (Krebs) cycle: Mitochondrial matrix
Oxidative phosphorylation (electron transport chain + chemiosmosis): Inner mitochondrial membrane (cristae)
Why: The light reactions need membrane-embedded photosystems, so they occur on the thylakoid membranes; the Calvin cycle's enzymes (including rubisco) are dissolved in the stroma. Glycolysis needs no organelle and occurs in the cytoplasm. Pyruvate oxidation and the citric acid cycle occur in the mitochondrial matrix. The ETC and ATP synthase are embedded in the inner mitochondrial membrane, whose folds (cristae) increase surface area.

B. Which organelles?

"A cell in one of the greenhouse leaves contains:"
Answer: C. Both chloroplasts and mitochondria, because plants photosynthesize and also carry out cellular respiration
Why: Autotrophs make their own glucose, but they still have to break it down to make ATP for work such as active transport and growth. Plant cells therefore contain both organelles. A is the most common misconception.
Hints (each costs 5% atmosphere):
(1) Photosynthesis happens entirely inside the chloroplast. The light reactions need membrane-bound photosystems; the Calvin cycle's enzymes float in fluid.
(2) Glycolysis doesn't need any organelle at all. The electron transport chain is made of proteins embedded in a membrane, and the membrane folded into cristae has the most surface area.
Data log entry unlocked: Every stage has an address
Chloroplast: light-dependent reactions on the thylakoid membranes, Calvin cycle in the stroma.
Cellular respiration: glycolysis in the cytoplasm, then pyruvate oxidation and the citric acid cycle in the mitochondrial matrix, then oxidative phosphorylation on the inner mitochondrial membrane.
Plant cells contain both organelles: autotrophs respire too.

Room 3: Solar Array
Light-dependent reactions. Location: Thylakoid membranes.
HELIX's message: Grow lights are back on, but the array's sorting system is scrambled. Tell it what the light reactions take in, what they give off, and what they never touch.

A. Sort the light-reaction molecules

"Mark each item as an input, an output, or not used by the light-dependent reactions."
Answer:
Input: Light energy, H_2O, $NADP^+$, ADP + Pi
Output: O_2, ATP, NADPH
Not used: CO_2, Glucose
Why: Inputs: light energy, H_2O, $NADP^+$, ADP + Pi. Outputs: O_2, ATP, NADPH. Not used: CO_2 and glucose, which belong to the Calvin cycle.

B. Trace the oxygen

"Biodome's sensors detect O2 coming from the plants. Where do those oxygen atoms come from?"

Answer: C. H2O, split at photosystem II to replace the electrons that light energized

Why: Photolysis at photosystem II splits water into electrons, H+, and O2. Isotope-tracing experiments with 18O confirmed that the released oxygen comes from water, not CO2 (A, a classic misconception). The oxygen in CO2 ends up in glucose and water.

C. How the array makes ATP

"How does the thylakoid membrane actually make ATP?"

Answer: B. The electron transport chain pumps H+ into the thylakoid lumen; H+ flows back out to the stroma through ATP synthase, which phosphorylates ADP

Why: This is chemiosmosis (photophosphorylation). Energized electrons moving down the ETC power H+ pumping into the lumen, so the lumen has a low pH. H+ diffuses back to the stroma through ATP synthase, so ATP is made on the stroma side, right where the Calvin cycle needs it. C reverses the gradient.

Hints (each costs 5% atmosphere):

(1) The light reactions make the energy carriers the Calvin cycle spends, but they never touch carbon.

(2) Photosystem II takes electrons from water. H+ piles up inside the thylakoid (the lumen) and then flows down its gradient back out.

Data log entry unlocked: Water is split; light becomes ATP and NADPH

Chlorophyll in photosystems II and I absorbs light (mostly red and blue wavelengths) and energizes electrons.

Water is split at photosystem II (photolysis), which is where the O2 comes from. The electron transport chain pumps H+ into the thylakoid lumen; H+ flows back to the stroma through ATP synthase (chemiosmosis). Electrons end on NADP+, forming NADPH.

Products: ATP and NADPH go to the Calvin cycle, and O2 leaves the leaf.

Room 4: Greenhouse Lab

The Calvin cycle. Location: Stroma.

HELIX's message: The greenhouse is absorbing light but making no sugar. Feed the Calvin cycle controller the right inputs, the right combination, and the right sequence.

A. Sort the Calvin cycle molecules

"Mark each item as an input, an output, or not used directly by the Calvin cycle."

Answer:

Input: CO2, ATP, NADPH

Output: G3P (used to build glucose), NADP+, ADP + Pi

Not used directly: O2, Light energy

Why: Inputs: CO2, ATP, NADPH. Outputs: G3P (two G3P make one glucose), ADP + Pi, NADP+. Not used directly: O2 and light. The cycle depends on light only indirectly, through the ATP and NADPH it receives.

B. Crack the controller combination

“One glucose molecule is built from 2 G3P, which takes 6 turns of the Calvin cycle. Enter what those 6 turns use.”

Answer:

CO_2 fixed: 6

ATP used: 18

NADPH used: 12

Why: Each turn fixes 1 CO_2 and spends 3 ATP and 2 NADPH, so 6 turns use 6 CO_2, 18 ATP, and 12 NADPH to make 2 G3P (one glucose). Combination: 6, 18, 12.

C. Restart the cycle in order

“Arrange the three phases in the order they happen, starting with carbon entering the cycle.”

Answer:

1. Carbon fixation: rubisco attaches CO_2 to RuBP (a 5-carbon sugar), forming 3-PGA
2. Reduction: ATP and NADPH convert 3-PGA into G3P (a 3-carbon sugar)
3. Regeneration: more ATP rebuilds RuBP from G3P so the cycle can turn again

Why: Carbon fixation (rubisco attaches CO_2 to RuBP, forming 3-PGA), then reduction (ATP + NADPH turn 3-PGA into G3P), then regeneration of RuBP (uses more ATP).

D. Lights out

“At night the grow lights shut off. Within minutes the Calvin cycle stops, even though it’s called ‘light-independent.’ Why?”

Answer: C. It runs out of the ATP and NADPH that only the light reactions supply

Why: The two stages are coupled: the Calvin cycle spends ATP and NADPH and returns ADP + Pi and NADP+ to the thylakoids to be recharged. Without light, that supply stops within minutes, so “light-independent” does not mean it runs in the dark for long.

Hints (each costs 5% atmosphere):

(1) Per turn: 1 CO_2 in, 3 ATP and 2 NADPH spent. Multiply by 6.

(2) The Calvin cycle runs on the products of the light reactions and sends the “empty” carriers (ADP, NADP+) back to be recharged. Fixation comes first, and RuBP has to be regenerated last.

Data log entry unlocked: Carbon from the air becomes sugar

In the stroma, rubisco fixes CO_2 onto RuBP. ATP and NADPH from the light reactions reduce the product to G3P, and more ATP regenerates RuBP.

Six turns use 6 CO_2, 18 ATP, and 12 NADPH to make 2 G3P, which is one glucose. The carbon in glucose comes from CO_2 in the air.

ADP, Pi, and NADP+ return to the thylakoids to be recharged, so the two stages depend on each other.

Room 5: Power Core

Cellular respiration. Location: Cytoplasm and mitochondrion.

HELIX’s message: Every cell in this dome, crew and plants alike, runs on ATP, and the generators are offline. Rebuild the respiration pipeline stage by stage.

A. Wire the respiration pipeline

“For each stage, choose its main input and its outputs per glucose molecule. One output set is a decoy.”

Answer:
Glycolysis: input Glucose (+ 2 ATP invested, 2 NAD+); outputs 2 pyruvate, 2 ATP (net), 2 NADH
Pyruvate oxidation: input 2 pyruvate; outputs 2 acetyl-CoA, 2 CO_2, 2 NADH
Citric acid (Krebs) cycle: input 2 acetyl-CoA; outputs 4 CO_2, 6 NADH, 2 $FADH_2$, 2 ATP
Oxidative phosphorylation: input NADH, $FADH_2$, and O_2; outputs About 26–28 ATP and 6 H_2O; NAD+ and FAD recycled
Why: Glycolysis: glucose → 2 pyruvate, net 2 ATP, 2 NADH. Pyruvate oxidation: 2 pyruvate → 2 acetyl-CoA, 2 CO_2, 2 NADH. Citric acid cycle (2 turns): 2 acetyl-CoA → 4 CO_2, 6 NADH, 2 $FADH_2$, 2 ATP. Oxidative phosphorylation: NADH, $FADH_2$, O_2 → about 26–28 ATP, 6 H_2O, with NAD+ and FAD recycled. Total ≈ 30–32 ATP per glucose. The decoy describes the Calvin cycle and light reactions.

B. Track the carbon

"Glucose enters with 6 carbon atoms. The CO_2 scrubbers log where each carbon leaves as CO_2. Enter the number released in each stage, per glucose."

Answer:
Glycolysis: 0
Pyruvate oxidation: 2
Citric acid cycle: 4
Why: Combination 0, 2, 4. Glycolysis splits glucose (6 C) into 2 pyruvate (3 C each) without releasing CO_2. Pyruvate oxidation removes one carbon from each pyruvate (2 CO_2). The citric acid cycle releases the remaining 4 carbons (2 per acetyl-CoA).

C. The oxygen's job

"The crew inhales the O_2 the plants released. What does that O_2 actually do inside a mitochondrion?"

Answer: A. It is the final electron acceptor at the end of the ETC, combining with electrons and H+ to form water
Why: O_2 is the final electron acceptor; it is reduced to H_2O. Without it, the ETC backs up, NAD+ is not regenerated, and ATP production collapses. B is a classic misconception: the oxygen atoms in exhaled CO_2 come from glucose and water, not inhaled O_2. NADH and $FADH_2$ (not O_2) carry electrons (D).

D. Where the ATP comes from

"Where is most of the ATP from one glucose made, and how?"

Answer: D. Oxidative phosphorylation: the ETC pumps H+ from the matrix into the intermembrane space, and H+ flows back through ATP synthase
Why: About 26–28 of the ≈30–32 ATP come from chemiosmosis. Electrons from NADH and $FADH_2$ power H+ pumping from the matrix into the intermembrane space; H+ flows back into the matrix through ATP synthase. Glycolysis and the citric acid cycle make only 2 ATP each by substrate-level phosphorylation. C reverses the gradient.

Hints (each costs 5% atmosphere):
(1) Follow the carbon: glucose (6 C) → 2 pyruvate (3 C each) → 2 acetyl-CoA (2 C each), releasing 2 CO_2 → the citric acid cycle releases the remaining carbons. Glycolysis releases none.
(2) NADH and $FADH_2$ are electron shuttles that deliver energy to the ETC; O_2 waits at the very end of the chain. In mitochondria the H+ gradient builds up in the intermembrane space.

Data log entry unlocked: Most ATP comes from a proton gradient
Glycolysis (cytoplasm) splits glucose into 2 pyruvate for a net 2 ATP and 2 NADH. **Pyruvate oxidation** and the **citric acid cycle** (matrix) release all 6 carbons as CO_2 and load NADH and $FADH_2$.
In **oxidative phosphorylation** (inner membrane), the ETC uses those electrons to pump H^+ into the intermembrane space; H^+ flows back through ATP synthase, making about 26–28 of the roughly 30–32 ATP per glucose. O_2 is the final electron acceptor and becomes H_2O.

Room 6: Main Airlock
Closing the loop. Location: The whole dome.
HELIX's message: Final door. Prove that the chloroplast systems and the mitochondrial systems are one loop, and I will open the airlock.

A. Route every product to its next job

"Each product of one process becomes useful somewhere else. Each destination is used once, and two are decoys."
Answer:
O_2 released when the light reactions split water: Final electron acceptor in oxidative phosphorylation
Glucose built from Calvin cycle G3P: Starting fuel for glycolysis
CO_2 released by pyruvate oxidation and the citric acid cycle: Fixed by rubisco in the Calvin cycle
H_2O formed at the end of the mitochondrial ETC: Split at photosystem II to replace electrons
ATP made by cellular respiration: Powers cell work like active transport, movement, and building proteins
Why: O_2 → final electron acceptor. Glucose → fuel for glycolysis. CO_2 → fixed by rubisco. H_2O → can be split at photosystem II. ATP → powers cell work (its energy is spent, not recycled). Light and stomata options are decoys: energy cannot be turned back into light for reuse, and stomata are pores for gas exchange, not storage.

B. Speak the airlock passphrase

"Complete the rule that keeps Biodome Six alive."
Answer:
Matter (the C, H, and O atoms)…: cycles between the two processes
Energy…: flows one way through, leaving as heat
Why: Matter cycles; energy flows. Atoms are rearranged, never created or destroyed. Energy enters as light, is stored in glucose, is transferred to ATP, and is eventually lost as heat (second law of thermodynamics).

C. Thought experiment

"Suppose the dome's lights failed permanently, but the plants, the crew, and the air all stayed sealed inside. What would happen?"
Answer: C. Life would eventually stop: with no new light energy, glucose and O_2 would run out, because energy lost as heat can't be turned back into glucose

Why: Matter could still cycle, but energy cannot. Every transfer loses energy as heat, so a constant input of light is required. Respiration consumes O2 rather than producing it (B), and photosynthesis cannot run on heat (D).

D. File your report to HELIX

"In 2–4 sentences, explain how the products of photosynthesis are used in cellular respiration and how the products of cellular respiration can be used in photosynthesis. HELIX checks for all four key molecules and at least 25 words."

Answer: The game checks that the report mentions glucose, oxygen, carbon dioxide, and water in at least 25 words. Score the quality with the rubric in Section 3.

Why: Sample full-credit answer: "Photosynthesis produces glucose and O2. Cellular respiration uses that glucose as its fuel in glycolysis and uses the O2 as the final electron acceptor of the electron transport chain. Respiration releases CO2 and H2O, which a plant can take back in: CO2 is fixed by rubisco in the Calvin cycle and H2O is split in the light reactions. Matter cycles between the two processes, but energy flows through and is lost as heat." Score on the 4-point scale in the facilitator guide.

Hints (each costs 5% atmosphere):

(1) Look back at Room 1: the products of each equation are the reactants of the other.

(2) Energy is different from matter. Light energy can be stored and transferred, but every transfer loses some as heat, and heat can't be turned back into glucose.

Data log entry unlocked: Matter cycles, energy flows

Photosynthesis makes the glucose and O2 that cellular respiration uses; respiration releases the CO2 and H2O that photosynthesis can use. Those atoms cycle between **autotrophs** and **heterotrophs** through gas exchange at leaf stomata and lungs.

Energy flows one way: light → glucose → ATP → heat. That's why Biodome Six can recycle its air forever but needs its lights on.

3. Scoring the written report (Room 6 D)

The game will not accept the report until it mentions glucose, oxygen, carbon dioxide, and water in at least 25 words. Use this scale to judge the science.

Score	Description
4	Explains both directions correctly with all four molecules and names a specific use for at least two of them (for example, O2 as the ETC's final electron acceptor, CO2 fixed by rubisco in the Calvin cycle, water split at photosystem II, glucose as the fuel for glycolysis). Notes that energy is not recycled, or contains no errors at all.
3	Explains both directions correctly with all four molecules, but only in general terms (for example, "respiration uses the oxygen").
2	Explains one direction well, or both directions

	with one scientific error (for example, saying energy is recycled, or that plants do not respire).
1	Lists the molecules without explaining how one process uses the other's products.

Sample 4-point answer: "Photosynthesis produces glucose and O2. Cellular respiration uses that glucose as its fuel in glycolysis and uses the O2 as the final electron acceptor of the electron transport chain. Respiration releases CO2 and H2O, which a plant can take back in: CO2 is fixed by rubisco in the Calvin cycle and H2O is split in the light reactions. Matter cycles between the two processes, but energy flows through and is lost as heat."